Chapter 1

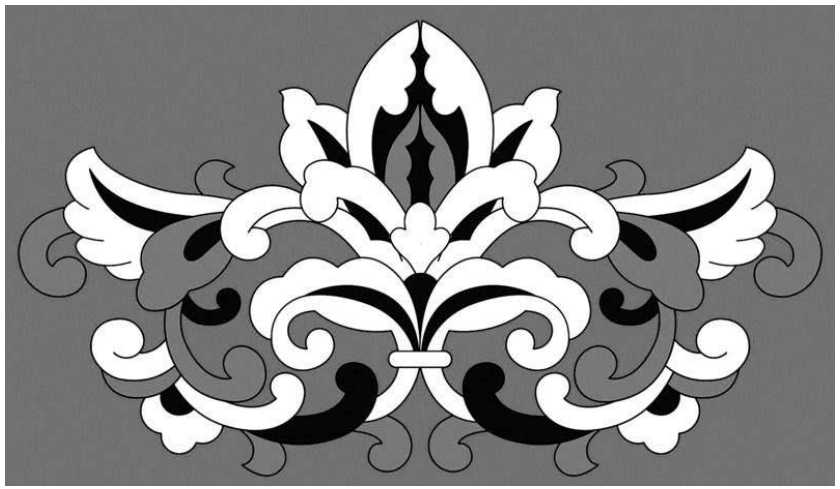
Quantum Theory

Part V

Contents

- Molecular Symmetry
- Representation of the Group

Symmetry







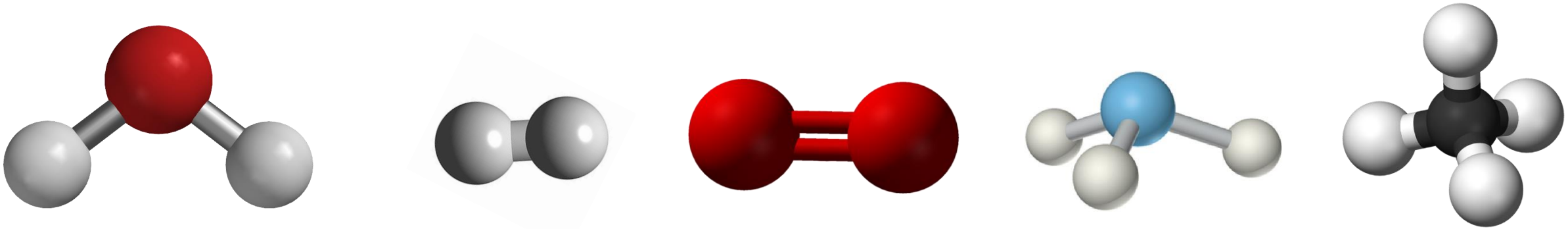








Molecular Symmetry



Classify molecular symmetry → **Classify molecular properties**

A rigorous understanding of symmetry explains fundamental observations in quantum chemistry, and in the applied areas of spectroscopy and crystallography.

Mathematics: An object is symmetric if there is a transformation that moves individual pieces of the object but doesn't change the overall shape.

Symmetry in physics has been generalized to mean invariance—that is, lack of change—under any kind of transformation

$$\left[\hat{A}, \hat{B}\right] = 0$$

Symmetry may be discussed systematically; the systematic discussion of symmetry is called **group theory**.

Symmetry operations and symmetry elements

A action that leaves an object looking the same after it has been carried out is called a **symmetry operation**.

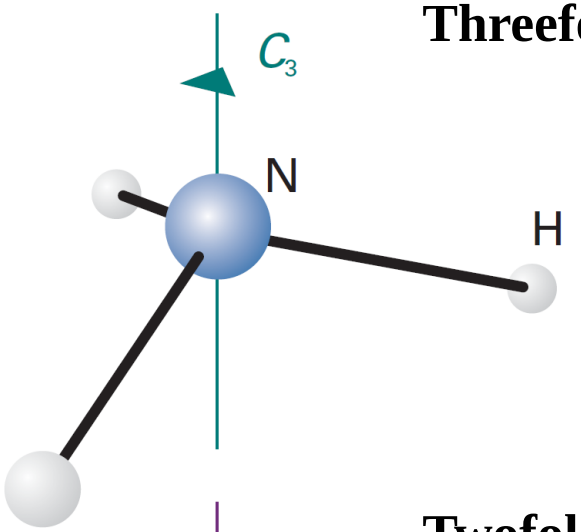
There is a corresponding **symmetry element** for each symmetry operation, which is the point, line, or plane with respect to which the symmetry operation is performed.

Symmetry operations:	Identity, *E*	Rotation	Reflection	Inversion
Symmetry elements:	entire object	a line	a plane	a point

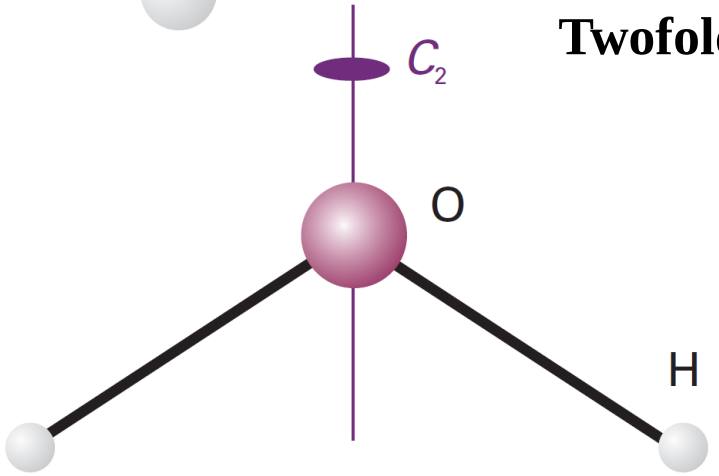
n-fold rotation C_n : A rotation through 360°/n

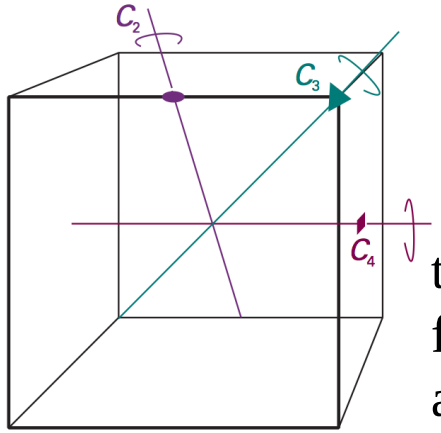
$$C_n^{\,k} = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \alpha = \frac{2k\pi}{n}$$

Threefold axis



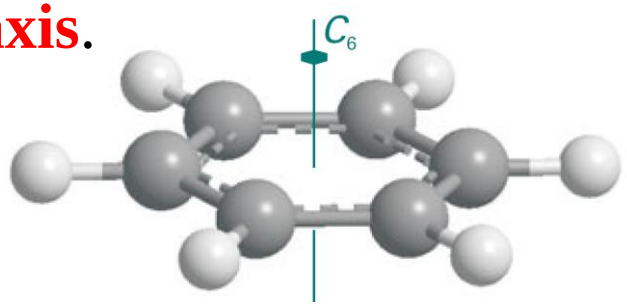
Twofold axis

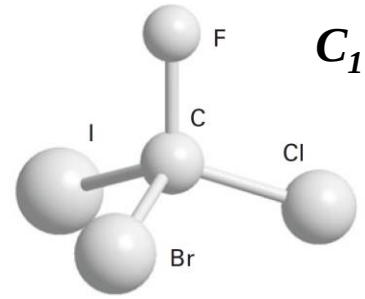




three C_4 axes,
four C_3 axes,
and six C_2 axes

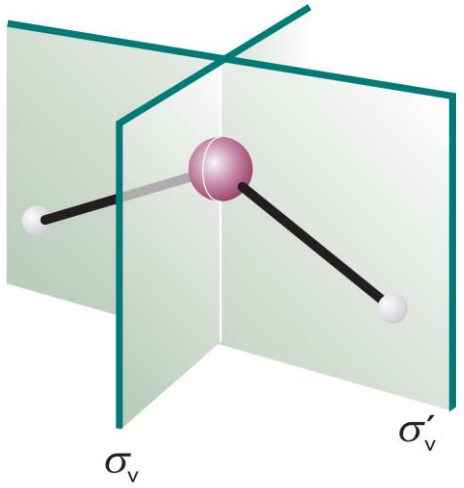
If a molecule possesses several rotation axes, then the one (or more) with the greatest value of n is called the **principal axis**.



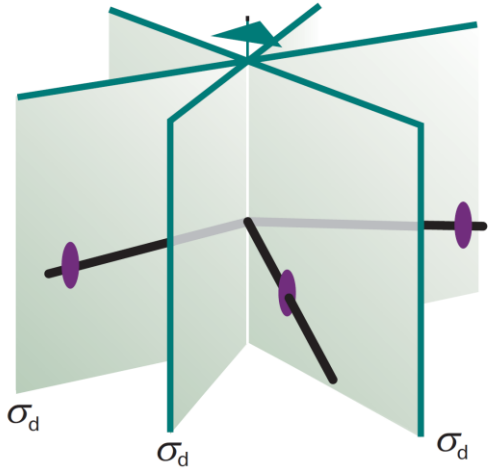


C_1

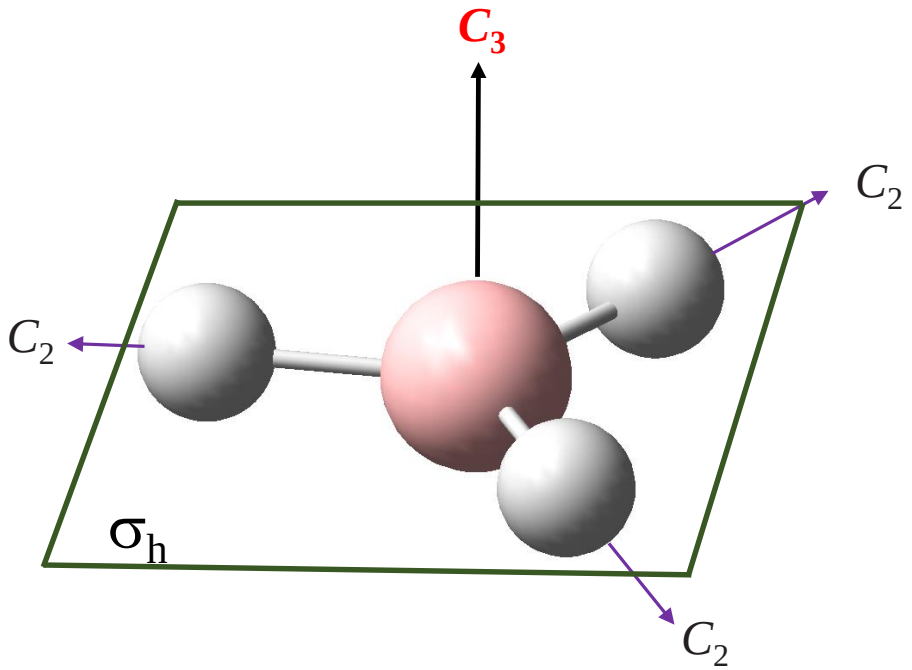
A **reflection** in a **mirror plane**, σ, may contain the principal axis of a molecule or be perpendicular to it.



Vertical mirror plane, σ_V, contains the principal axis



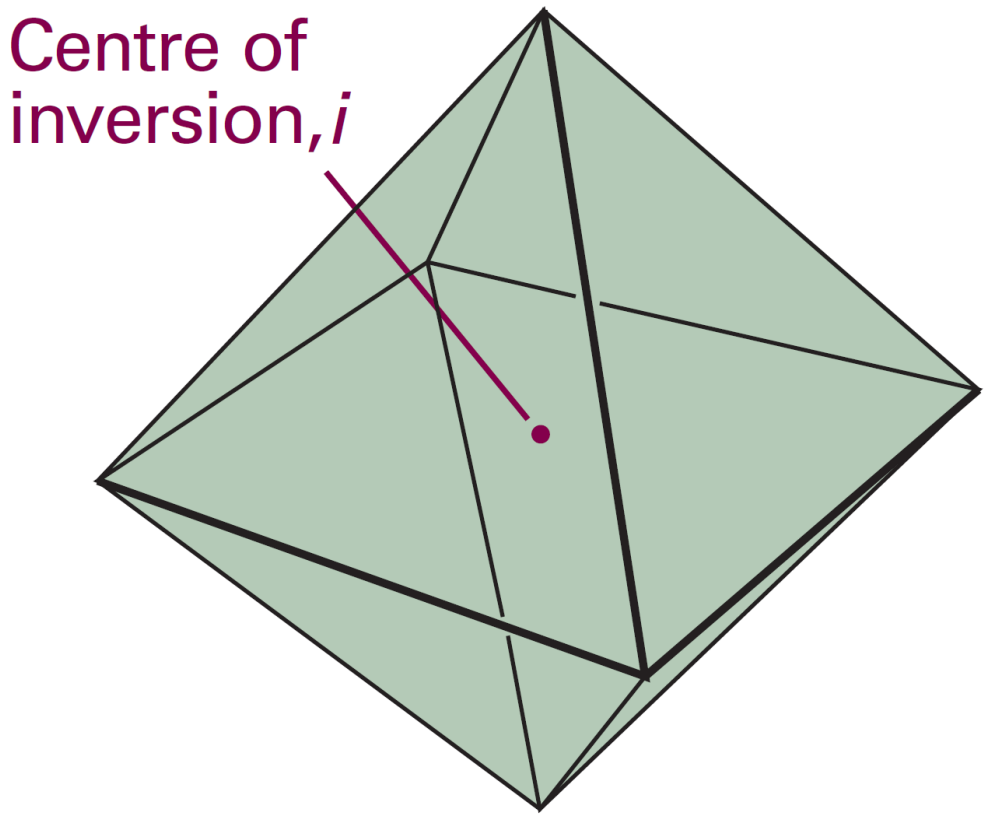
Dihedral mirror planes, σ_d, a vertical mirror plane that bisect the C_2 axes perpendicular to the principal axis.



Horizontal mirror planes, σ_h, perpendicular to the principal axis

Inversion through a **center of symmetry**, ***i***

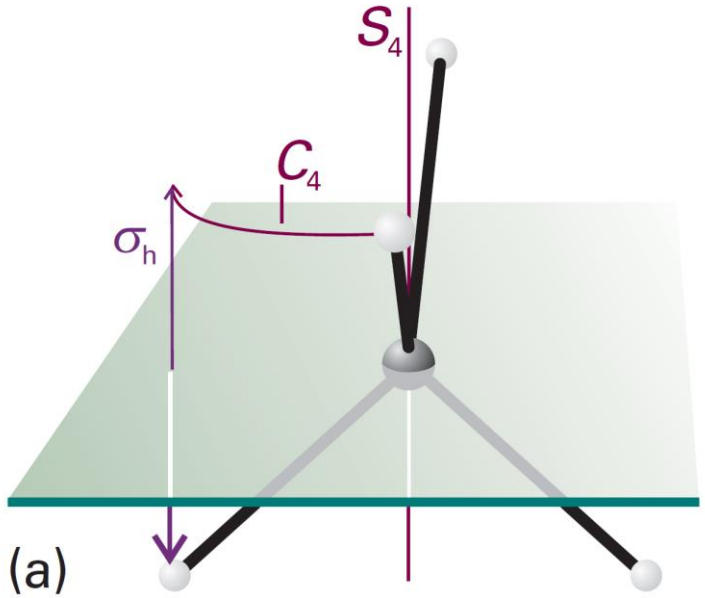
the point (x, y, z) $\rightarrow$ the point (-x, -y, -z)

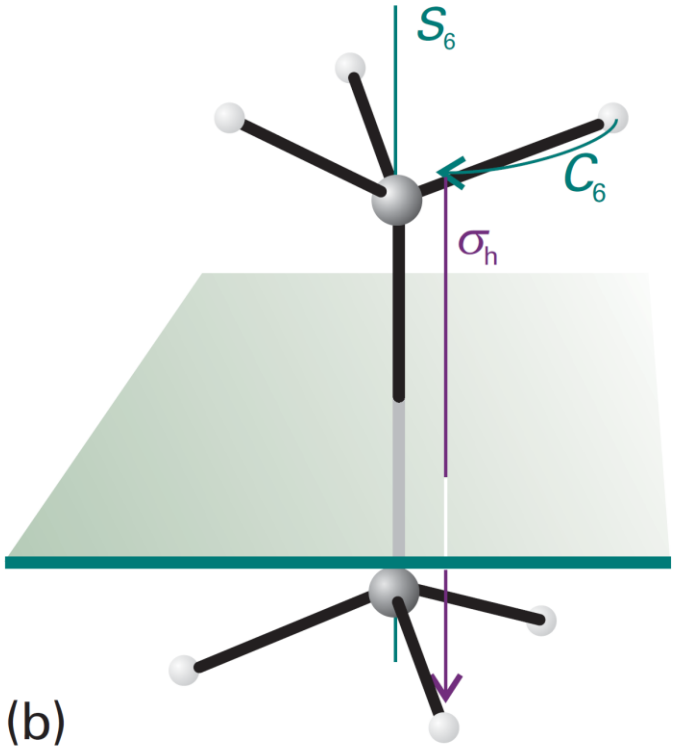


A regular octahedron has a center of inversion (*i*).

***n*-fold improper rotation** about an ***n*-fold improper rotation axis, S_n**

A rotation through 360°/*n* + a reflection through a plane perpendicular to the axis of that rotation: $S_n = \sigma_h C_n$





The symmetry classification of molecules

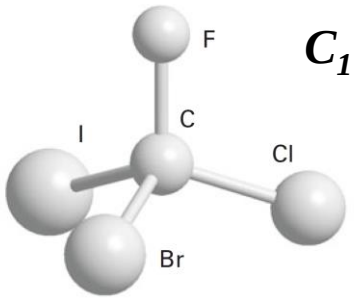
The classification of objects according to symmetry elements corresponding to operations that leave at least one common point unchanged gives rise to the **point groups (点群)**.

The groups

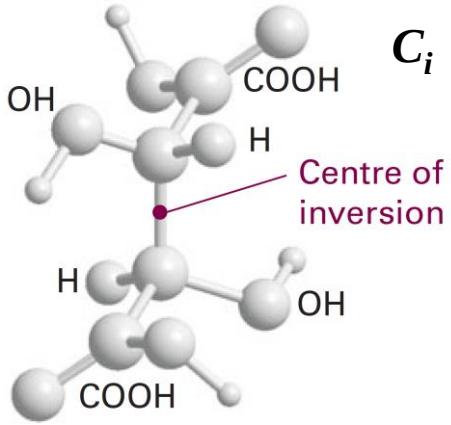
C_1, no element other than the identity E

C_i, has the identity E and the inversion alone

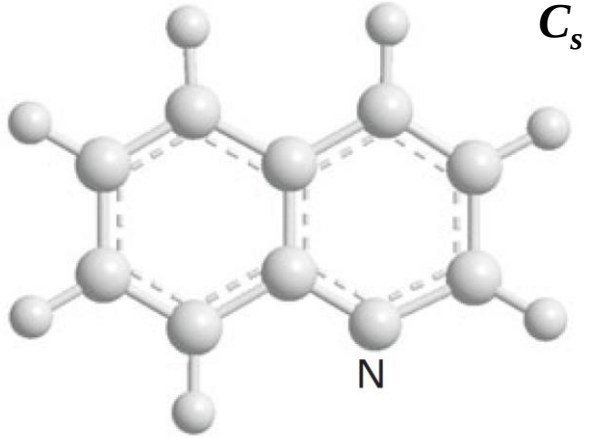
C_s, has the identity E and a mirror plane alone



C_1



C_i



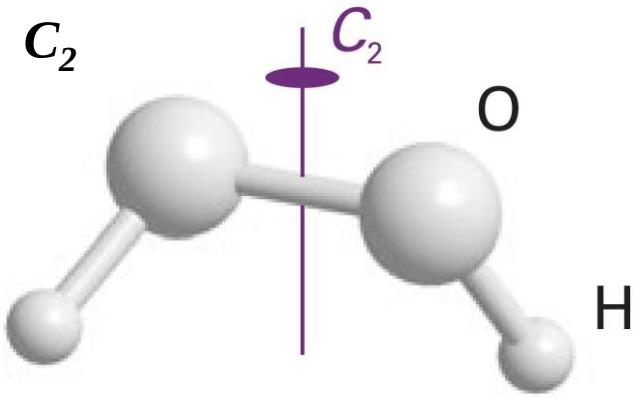
C_s

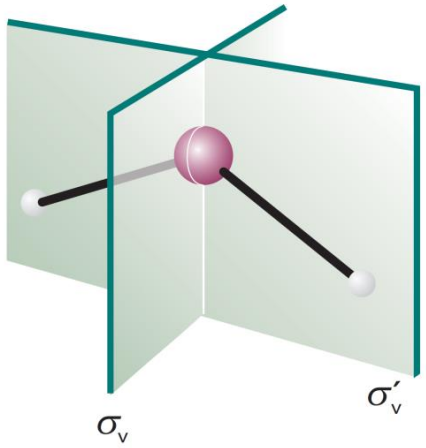
The groups

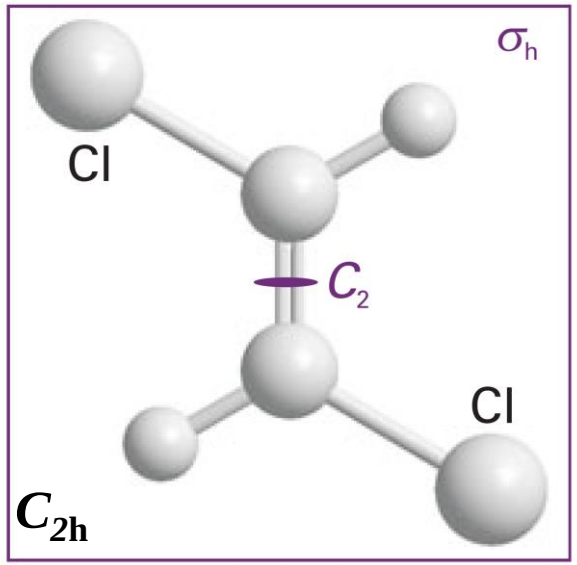
C_n, E + a C_n axis

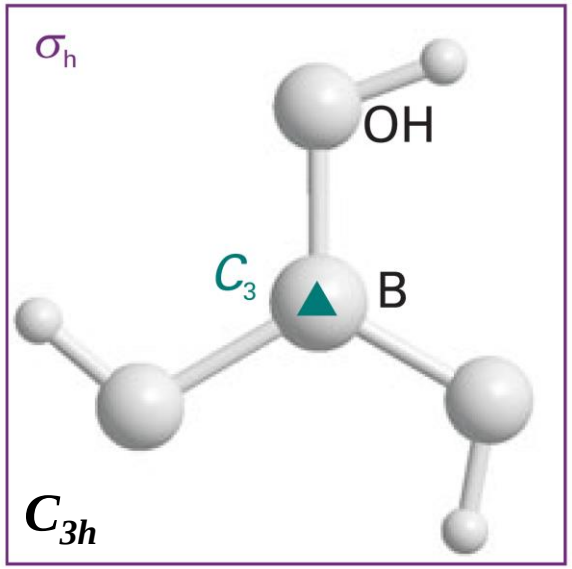
C_{nv}, E + a C_n axis + n σ_V

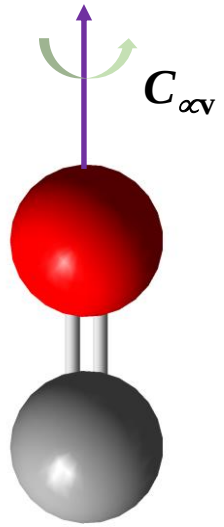
C_{nh}, E + a C_n axis + a σ_h









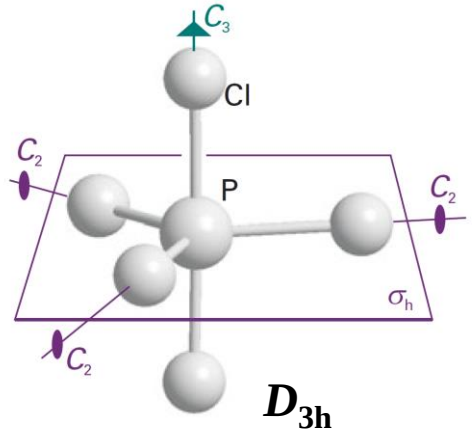


The groups

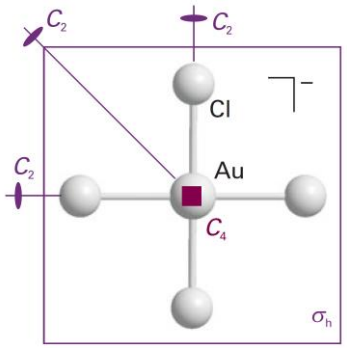
***D*$_n$**, E + a C_n principal axis + n C_2 axes $\perp$ C_n

***D*$_{nh}$**, E + a C_n principal axis + n C_2 axes $\perp$ C_n + a σ_h

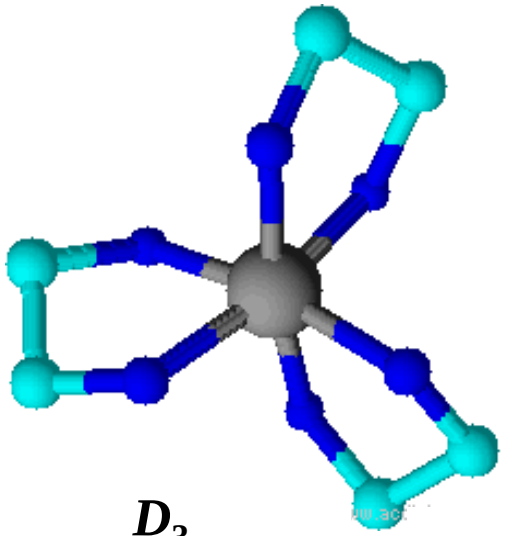
***D*$_{nd}$**, E + a C_n principal axis + n C_2 axes $\perp$ C_n + n σ_d



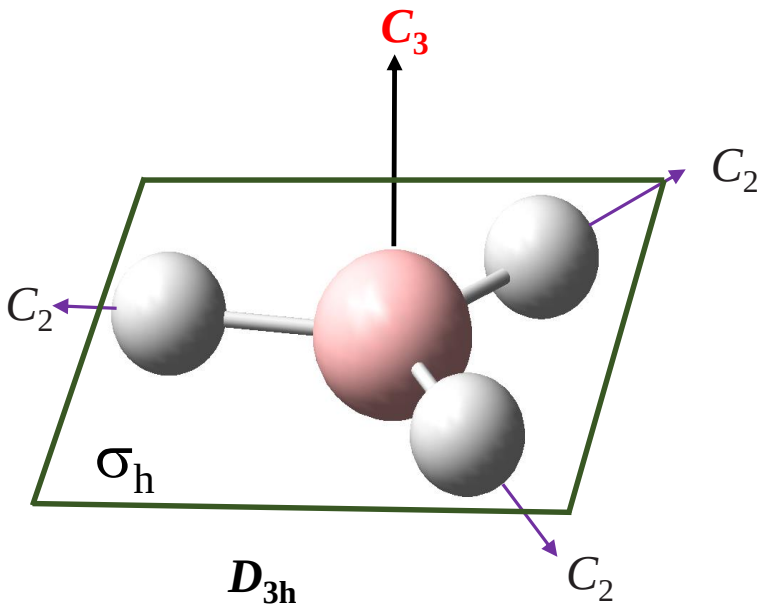
***D*$_{3h}$**



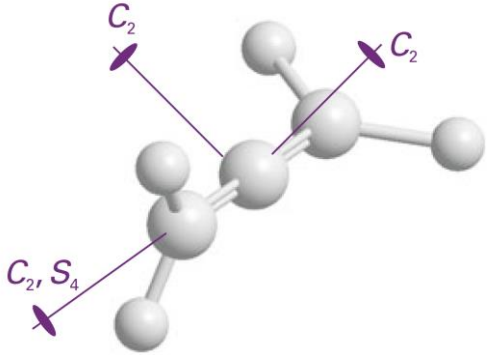
***D*$_{4h}$**



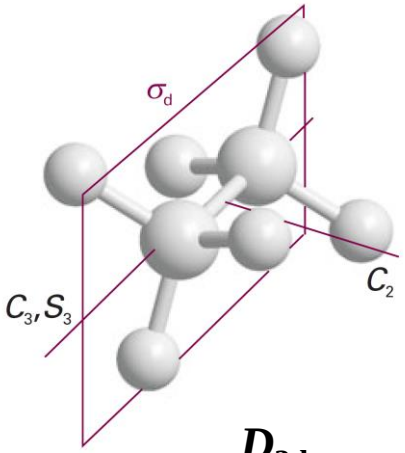
***D*$_3$**



***D*$_{3h}$**

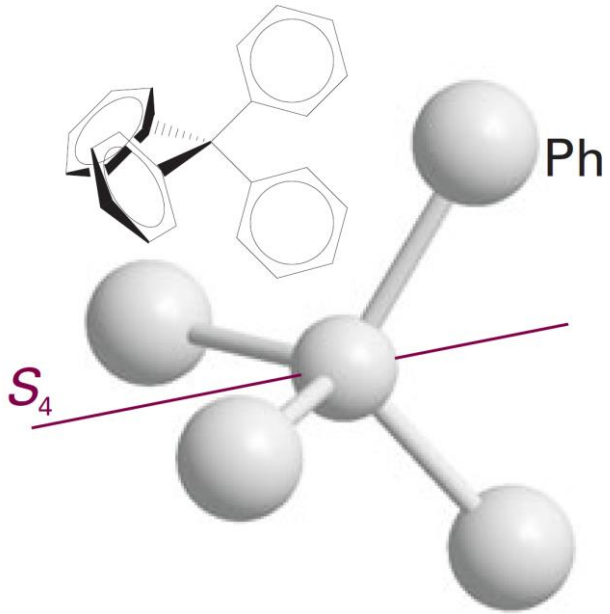


***D*$_{2d}$**

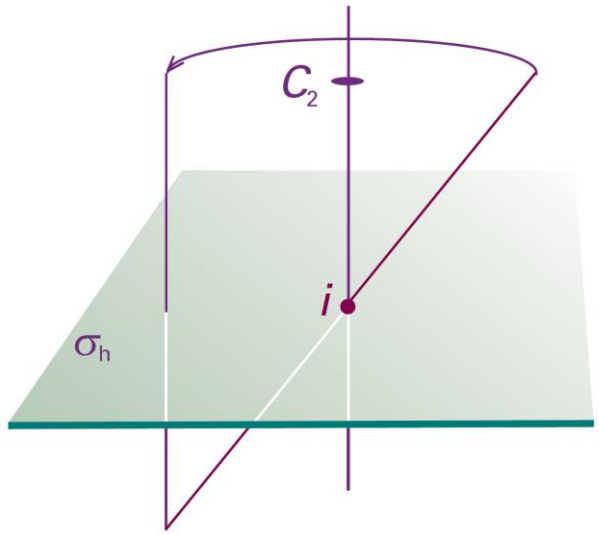


***D*$_{3d}$**

The groups S_n has **a** S_n improper rotation axis, $n=2m$, $m>1$



$$S_2 = \sigma_h C_2 = i$$



$S_1 = \sigma_h \rightarrow C_s$

$S_2 = i \rightarrow C_i$

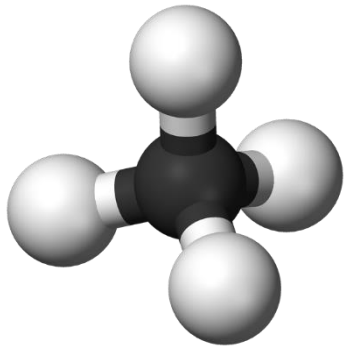
$S_3 = C_3 + \sigma_h \rightarrow C_{3h}$

$S_{2n+1} = C_{(2n+1)} + \sigma_h \rightarrow C_{(2n+1)h}$

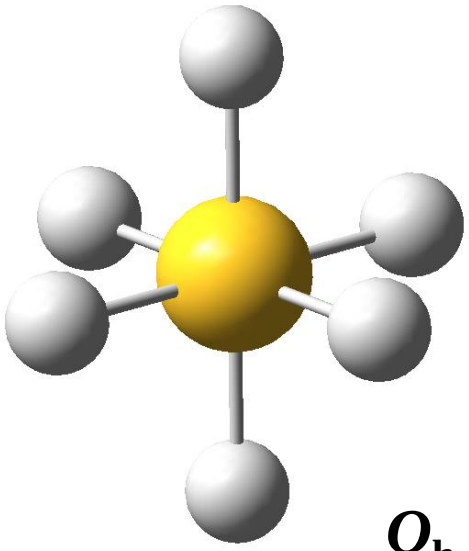
The cubic groups have more than one principal axis

Tetrahedral groups T, T_d, and T_h;
Octahedral groups O and O_h;
Icosahedral group, I

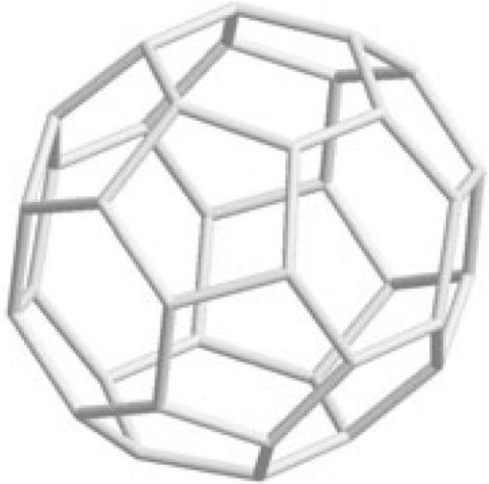
T: $4\,C_3, 3\,C_2$ $(T_h: T + 3\sigma_h)$ $(T_d: T + 3S_4)$
O: $4\,C_3, 3\,C_4$ $(O_h: O + 3\sigma_h)$
I: $6\,C_5, 10\,C_3$ $(I_h: I + i)$



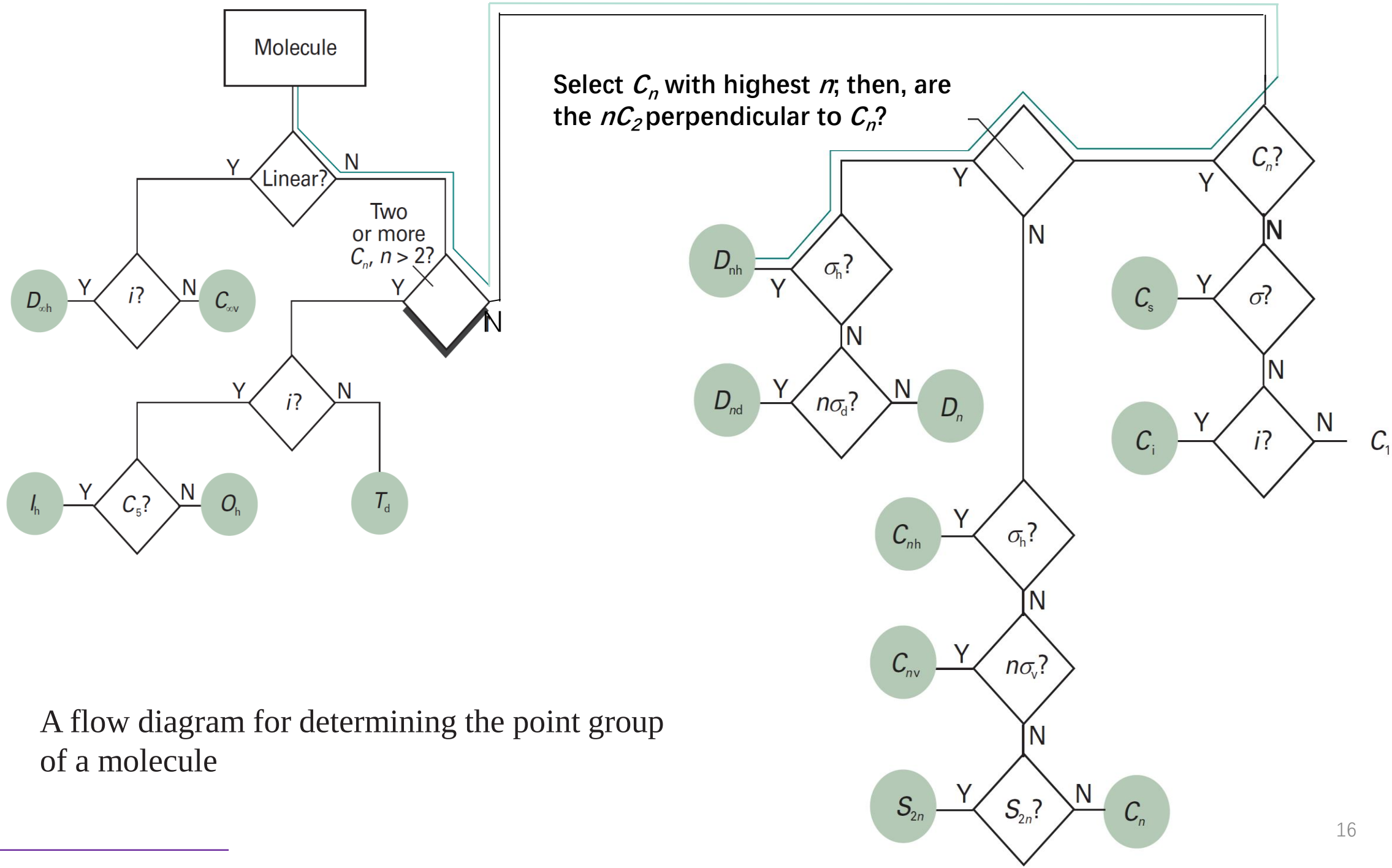
T_d



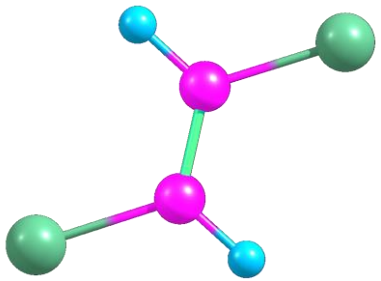
O_h



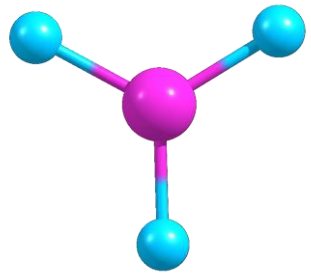
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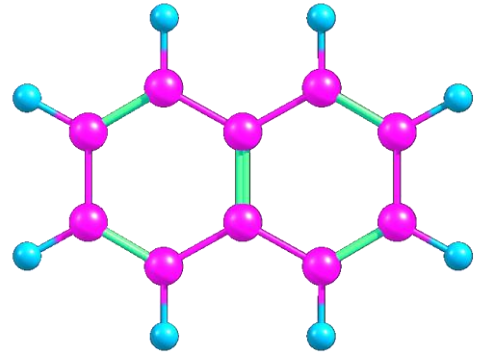
A flow diagram for determining the point group of a molecule



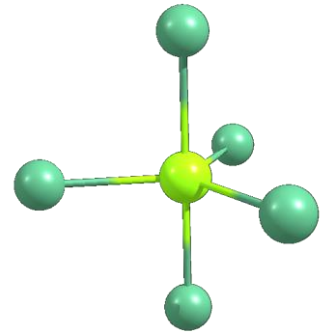
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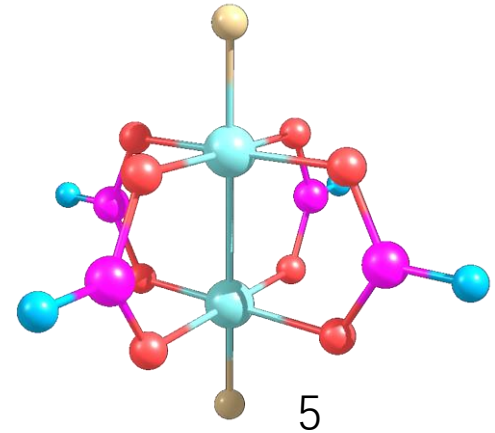
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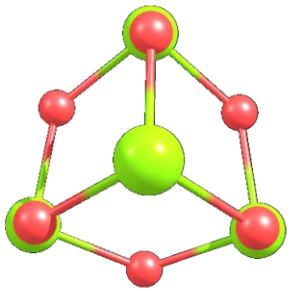
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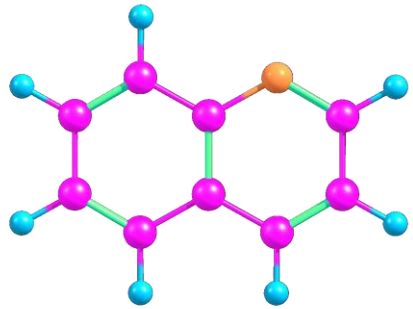
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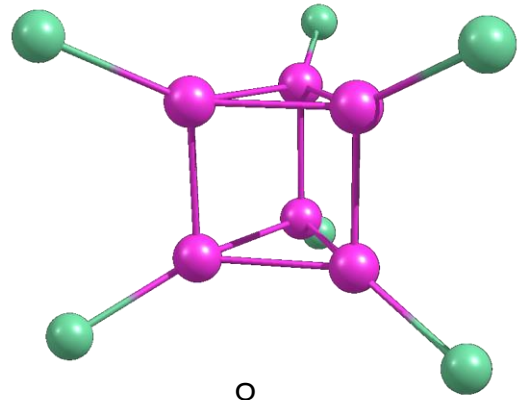
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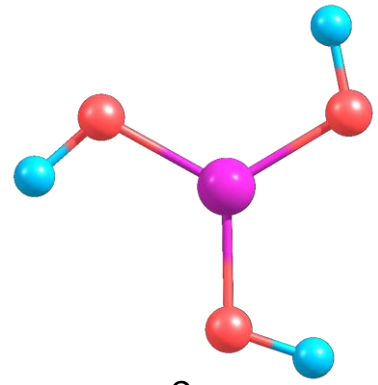
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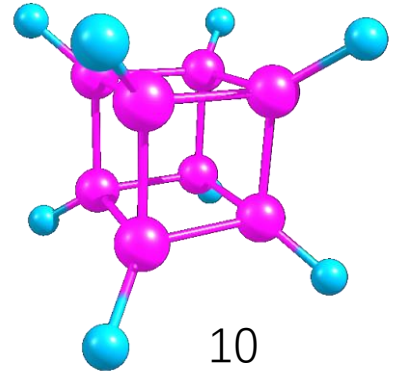
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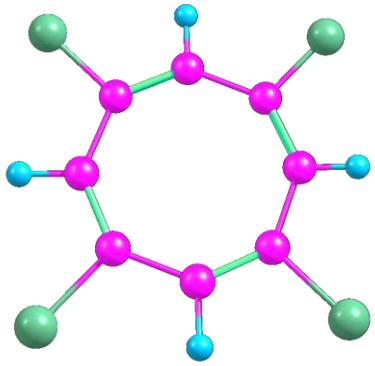
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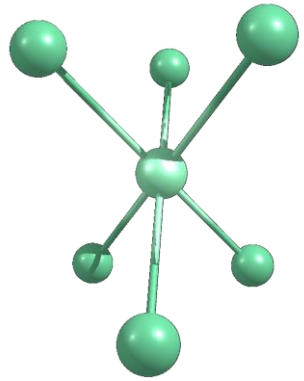
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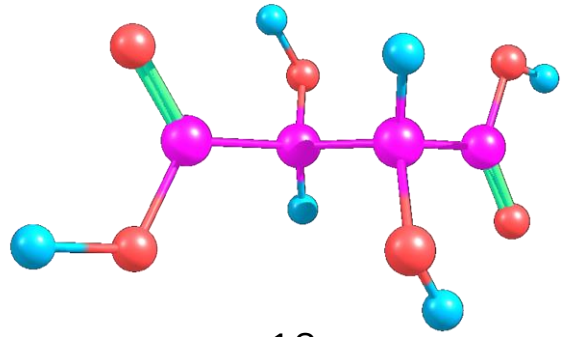
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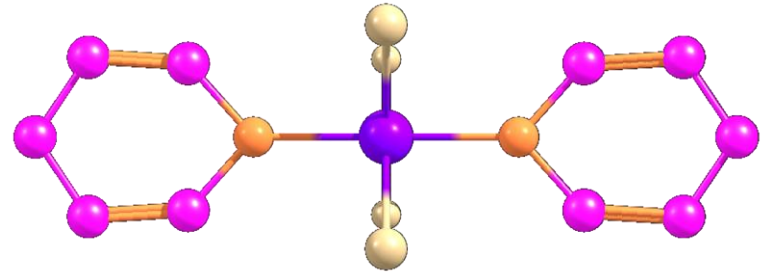
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